Data Analysis Report

Fake Job Postings Dataset

Generated: April 11, 2026

Total Records: 17,880 | Total Features: 18

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1. Executive Summary

This comprehensive report analyzes the Fake Job Postings dataset containing 17,880 job postings.
The dataset includes 18 features across binary, categorical, numeric, and text types.

Key Statistics:
- Total Records: 17,880
- Real Job Postings: 17,014 (95.157%)
- Fraudulent Job Postings: 866 (4.843%)
- Total Features Analyzed: 16

The analysis covers all feature types including 3 binary features,
8 categorical features, 0 numeric features,
and 5 text features.

2. Dataset Overview

Metric	Value
Total Records	**17,880**
Total Features	**18**
Real Job Postings	**17,014 (95.157%)**
Fraudulent Job Postings	**866 (4.843%)**
Binary Features	**3**
Categorical Features	**8**
Numeric Features	**0**
Text Features	**5**

3. Feature Classification

Binary Features:

telecommuting, has_company_logo, has_questions

Categorical Features:

location, department, salary_range, employment_type, required_experience, required_education, industry, function

Numeric Features:

Text Features:

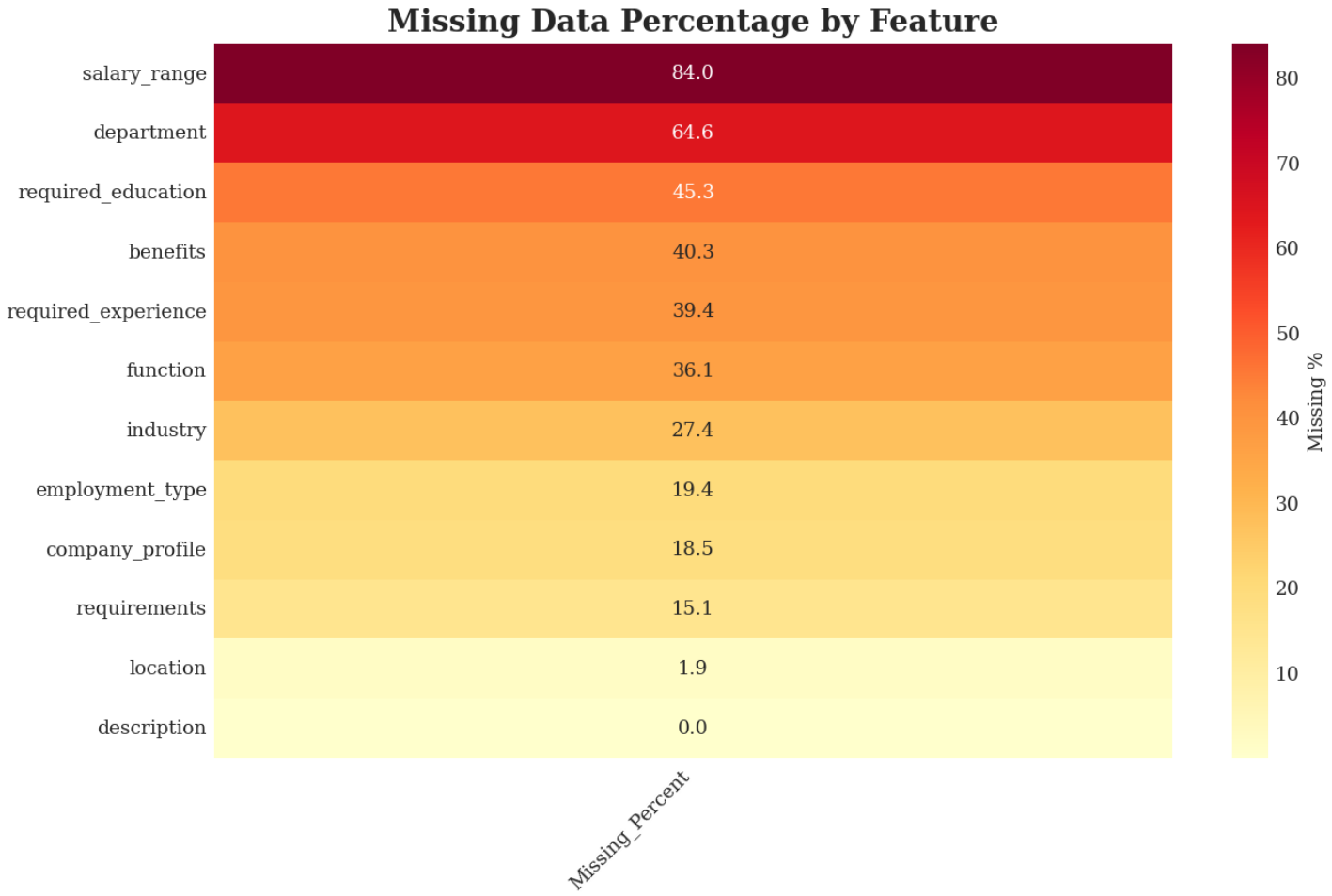
title, description, requirements, benefits, company_profile

4. Missing Data Analysis

Features with missing values (sorted by missing percentage):

Feature	Missing Values	Missing %
salary_range	15012	83.96%
department	11547	64.58%
required_education	8105	45.33%
benefits	7212	40.34%
required_experience	7050	39.43%
function	6455	36.1%
industry	4903	27.42%
employment_type	3471	19.41%
company_profile	3308	18.5%
requirements	2696	15.08%
location	346	1.94%
description	1	0.01%

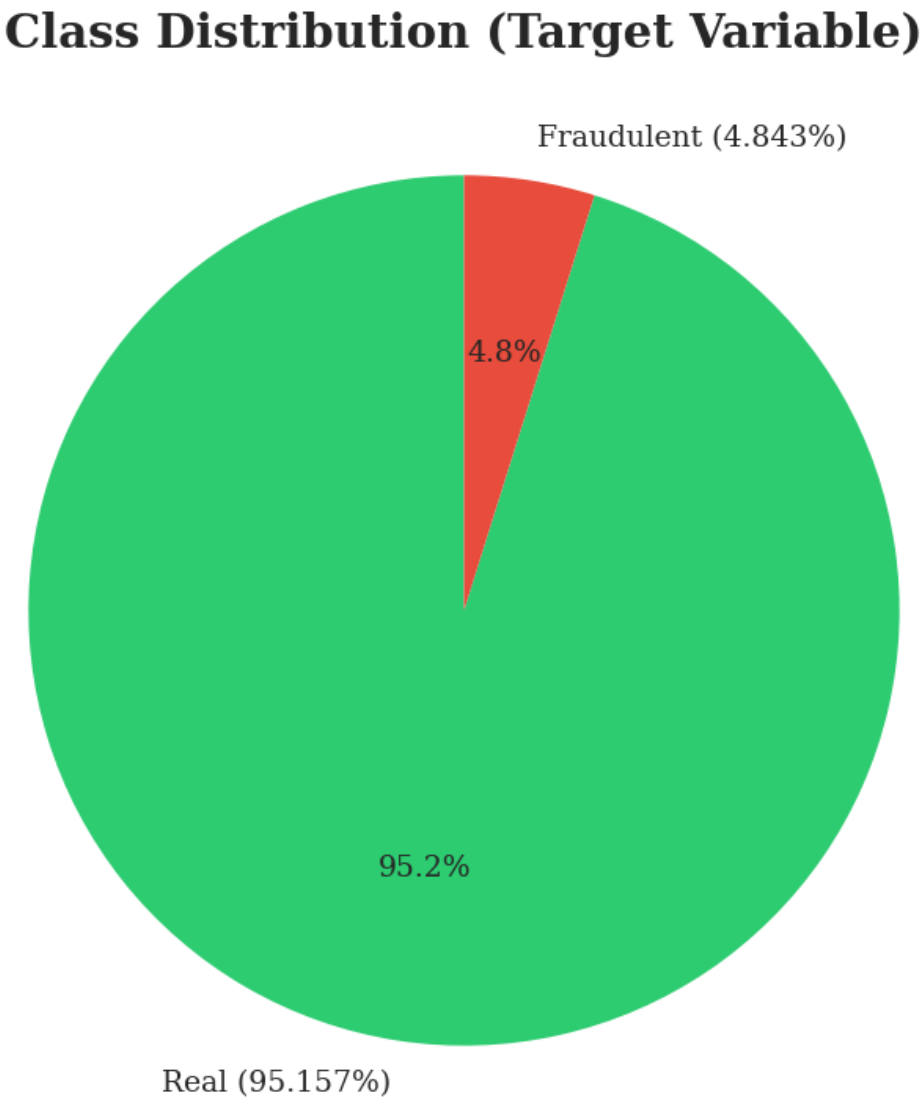
Missing Data Heatmap



5. Class Distribution

Target Variable: fraudulent
- Real Job Postings (0): 17,014 (95.157%)
- Fraudulent Job Postings (1): 866 (4.843%)

The dataset shows a class imbalance with 95.157% real postings and 4.843% fraudulent postings.



6. Binary Features Analysis

Feature: telecommuting

Correlation with target: +0.03452

P-value: 0.000004

Statistically Significant: Yes

Value Distribution:

- 0: 17113 (95.71%)
- 1: 767 (4.29%)

Feature: has_company_logo

Correlation with target: -0.26197

P-value: 0.000000

Statistically Significant: Yes

Value Distribution:

- 1: 14220 (79.53%)
- 0: 3660 (20.47%)

Feature: has_questions

Correlation with target: -0.09163

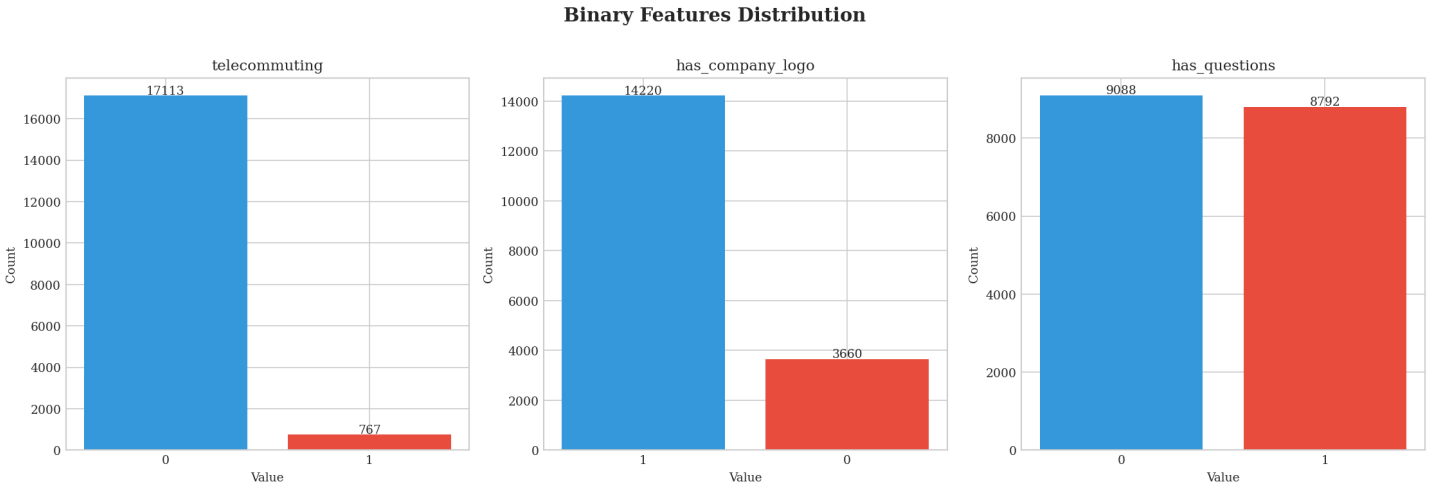
P-value: 0.000000

Statistically Significant: Yes

Value Distribution:

- 0: 9088 (50.83%)
- 1: 8792 (49.17%)

Binary Features Distribution



7. Categorical Features Analysis

Feature: location

Unique Categories: 3105

Mode: GB, LND, London

Cramér's V with target: 0.68530

Chi-square statistic: 8397.1279

P-value: 0.000000

Statistically Significant: Yes

Top 5 Categories:

- GB, LND, London: 718 (4.09%)
- US, NY, New York: 658 (3.75%)
- US, CA, San Francisco: 472 (2.69%)
- GR, I, Athens: 464 (2.65%)
- US, , : 339 (1.93%)

Feature: department

Unique Categories: 1337

Mode: Sales

Cramér's V with target: 0.47731

Chi-square statistic: 4073.4501

P-value: 0.000000

Statistically Significant: Yes

Top 5 Categories:

- Sales: 551 (8.7%)
- Engineering: 487 (7.69%)
- Marketing: 401 (6.33%)
- Operations: 270 (4.26%)
- IT: 225 (3.55%)

Feature: salary_range

Unique Categories: 874

Mode: 0-0

Cramér's V with target: 0.37223

Chi-square statistic: 2477.3919

P-value: 0.000000

Statistically Significant: Yes

Top 5 Categories:

- 0-0: 142 (4.95%)
- 40000-50000: 66 (2.3%)
- 30000-40000: 55 (1.92%)
- 25000-30000: 37 (1.29%)
- 45000-67000: 37 (1.29%)

Feature: employment_type

Unique Categories: 5

Mode: Full-time

Cramér's V with target: 0.07472

Chi-square statistic: 99.8286

P-value: 0.000000

Statistically Significant: Yes

Top 5 Categories:

- Full-time: 11620 (80.64%)
- Contract: 1524 (10.58%)
- Part-time: 797 (5.53%)
- Temporary: 241 (1.67%)
- Other: 227 (1.58%)

Feature: required_experience

Unique Categories: 7

Mode: Mid-Senior level

Cramér's V with target: 0.08417

Chi-square statistic: 126.6599

P-value: 0.000000

Statistically Significant: Yes

Top 5 Categories:

- Mid-Senior level: 3809 (35.17%)
- Entry level: 2697 (24.9%)
- Associate: 2297 (21.21%)
- Not Applicable: 1116 (10.3%)
- Director: 389 (3.59%)

Feature: required_education

Unique Categories: 13

Mode: Bachelor's Degree

Cramér's V with target: 0.16106

Chi-square statistic: 463.8083

P-value: 0.000000

Statistically Significant: Yes

Top 5 Categories:

- Bachelor's Degree: 5145 (52.63%)
- High School or equivalent: 2080 (21.28%)
- Unspecified: 1397 (14.29%)
- Master's Degree: 416 (4.26%)
- Associate Degree: 274 (2.8%)

Feature: industry

Unique Categories: 131

Mode: Information Technology and Services

Cramér's V with target: 0.30131

Chi-square statistic: 1623.2921

P-value: 0.000000

Statistically Significant: Yes

Top 5 Categories:

- Information Technology and Services: 1734 (13.36%)
- Computer Software: 1376 (10.6%)

- Internet: 1062 (8.18%)
- Marketing and Advertising: 828 (6.38%)
- Education Management: 822 (6.33%)

Feature: function

Unique Categories: 37

Mode: Information Technology

Cramér's V with target: 0.17487

Chi-square statistic: 546.7466

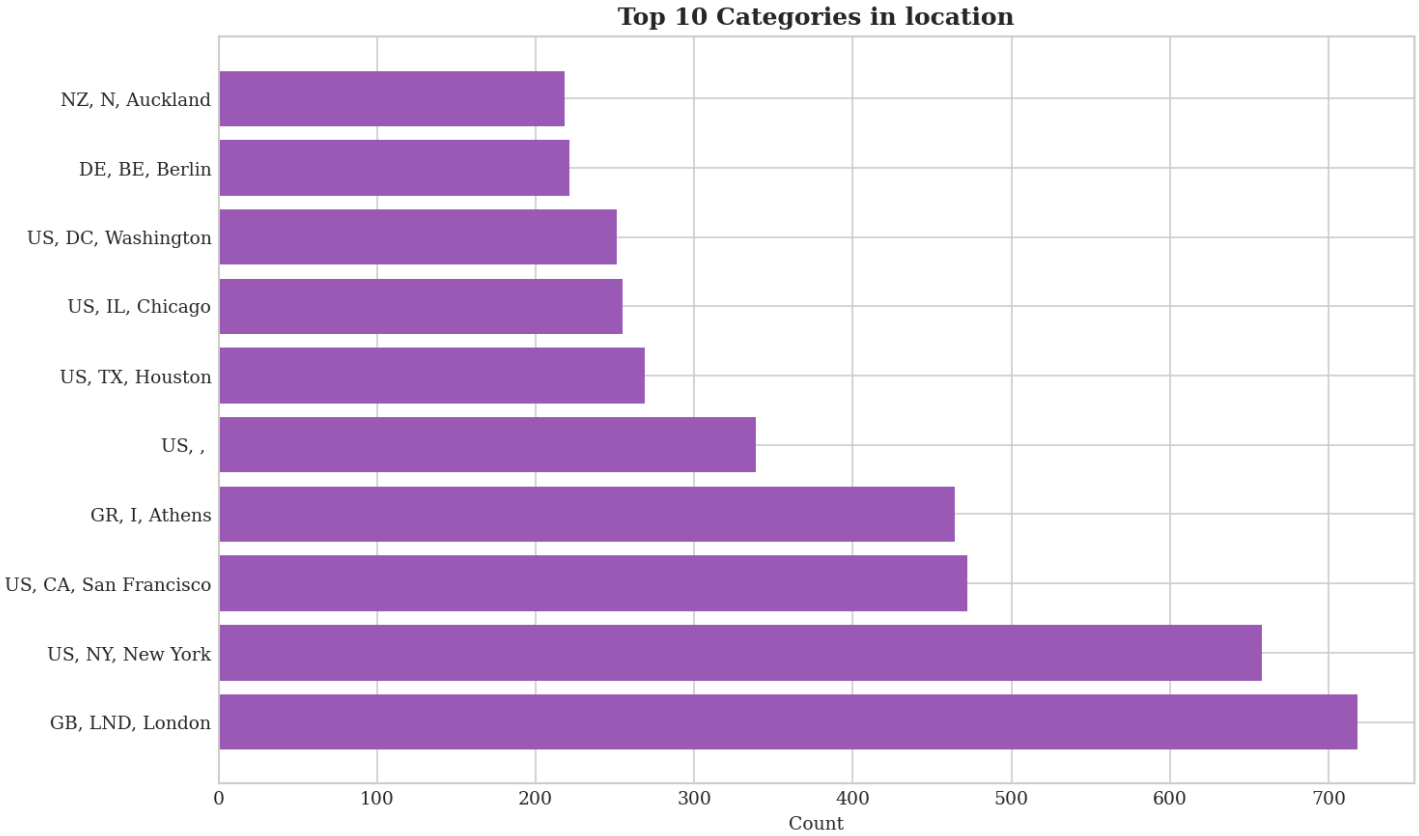
P-value: 0.000000

Statistically Significant: Yes

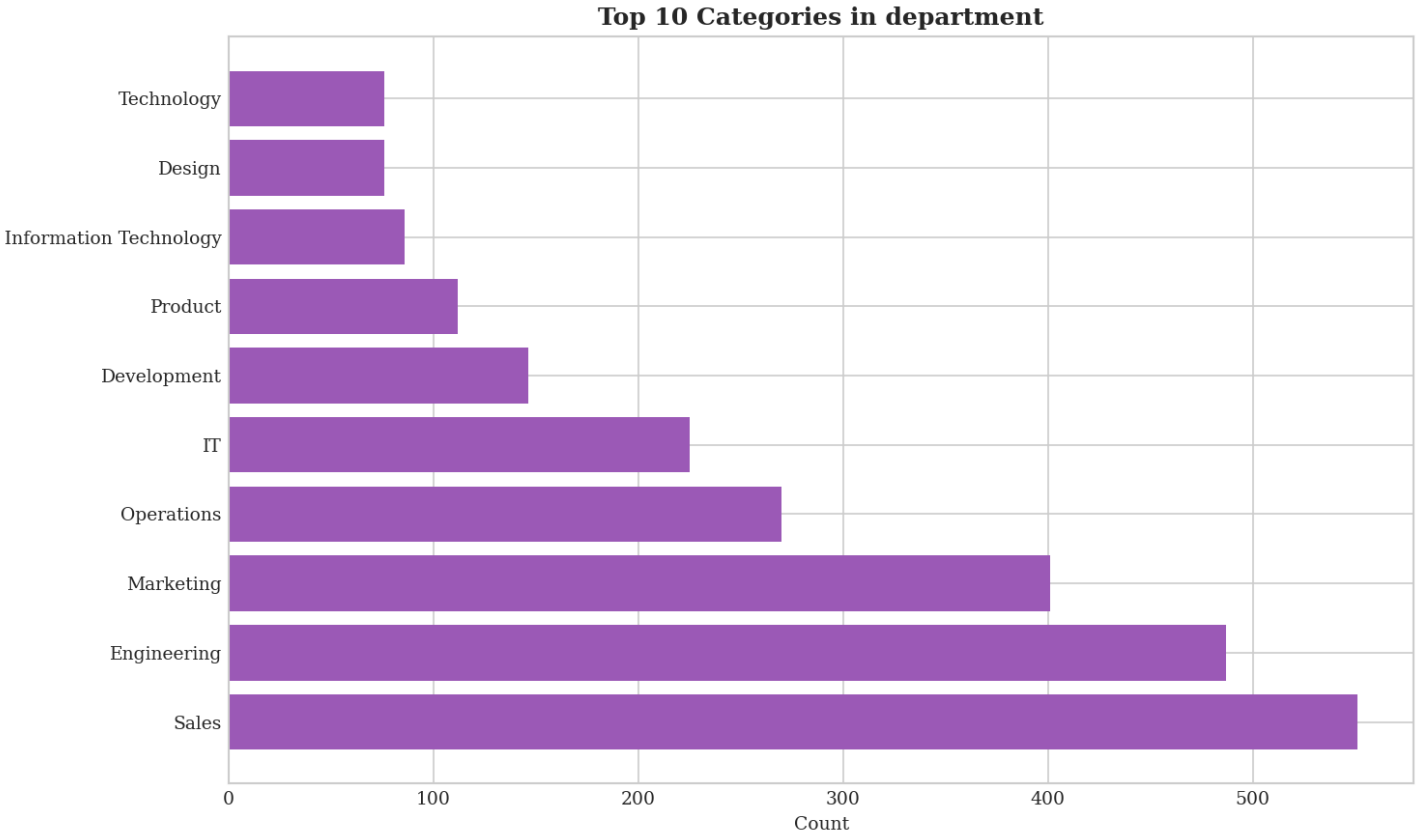
Top 5 Categories:

- Information Technology: 1749 (15.31%)
- Sales: 1468 (12.85%)
- Engineering: 1348 (11.8%)
- Customer Service: 1229 (10.76%)
- Marketing: 830 (7.26%)

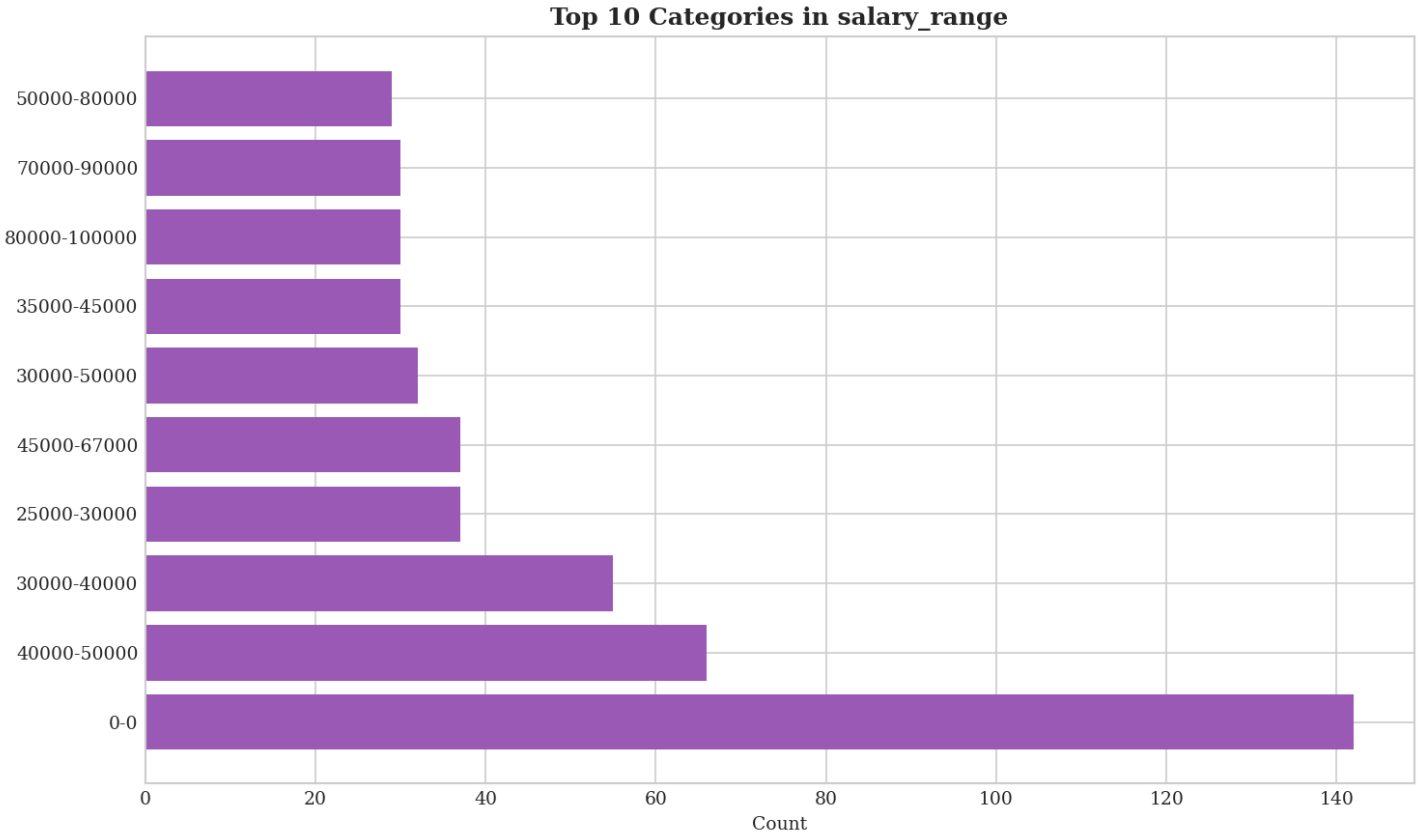
Top Categories: location



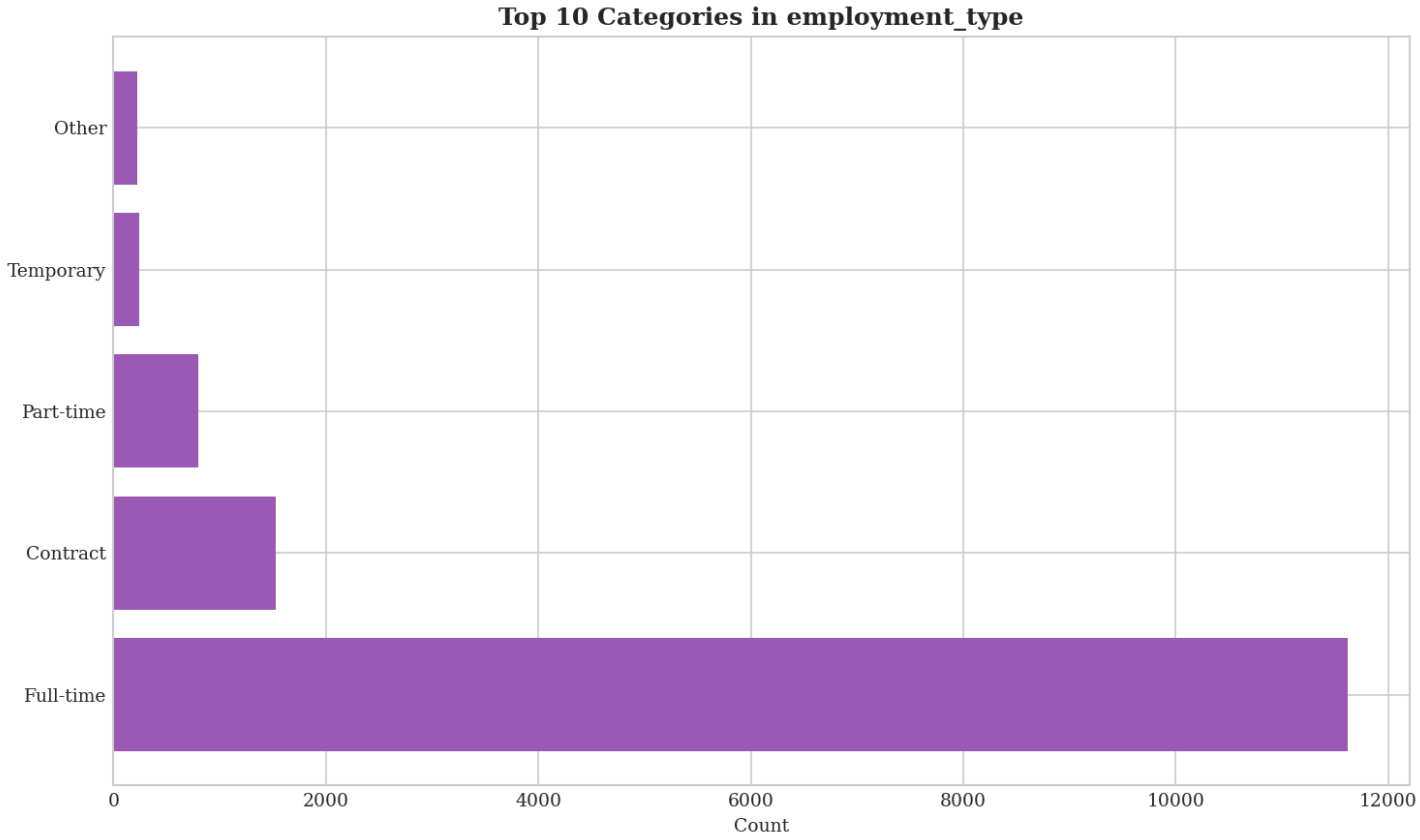
Top Categories: department



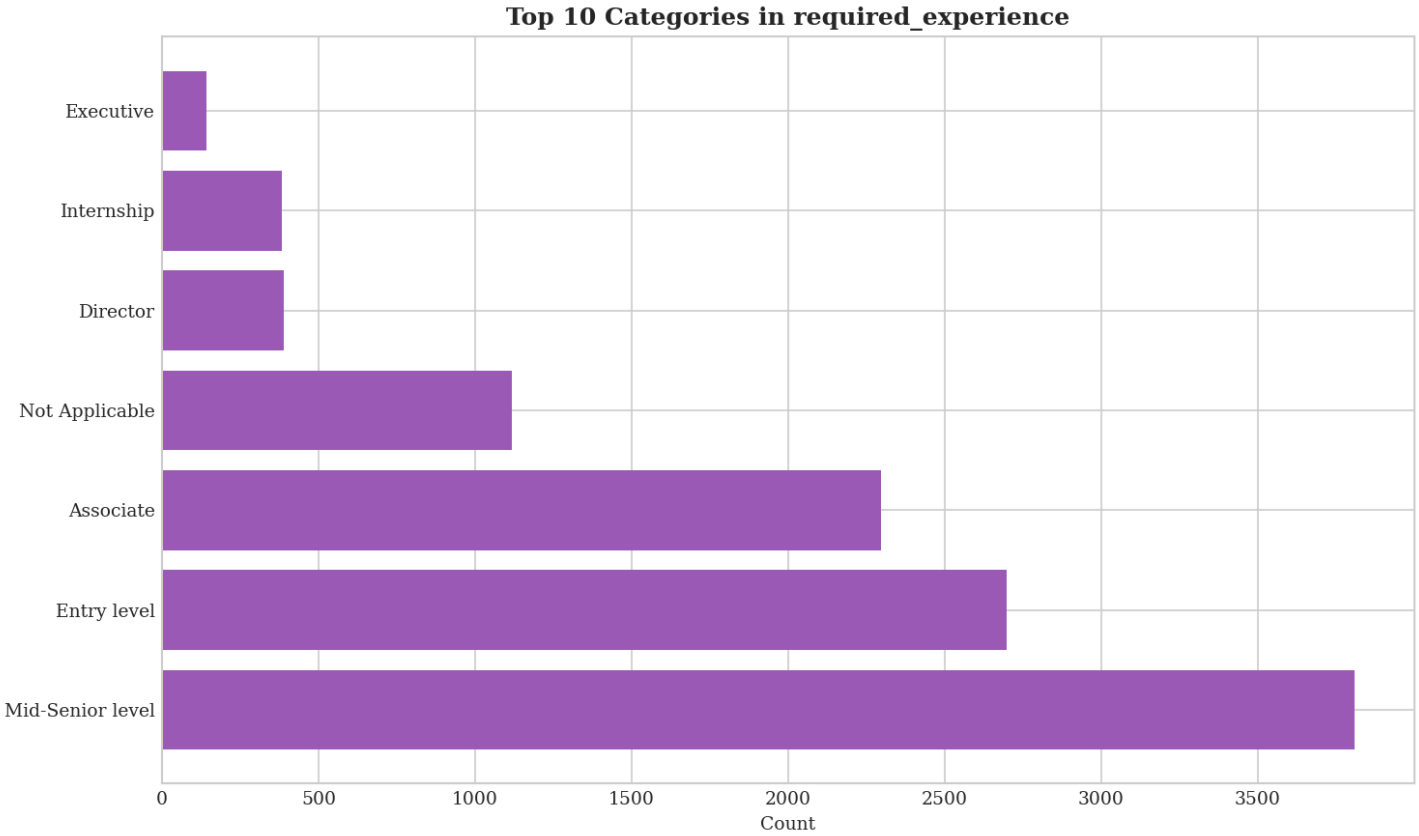
Top Categories: salary_range



Top Categories: employment_type



Top Categories: required_experience



8. Numeric Features Analysis

No numeric features found in the dataset.

All features in this dataset are either binary, categorical, or text.

9. Text Features Analysis

Feature: title

Character Length Statistics:

- mean: 28.53
- median: 25.0
- std: 13.87
- min: 3
- max: 142

Word Count Statistics:

- mean: 3.76
- median: 3.0
- std: 2.07
- min: 1
- max: 19

Average Length - Fraudulent: 30.67, Real: 28.42

Top 10 Most Common Words:

- 'manager': 2106 occurrences
- '-': 1951 occurrences
- 'developer': 1772 occurrences
- 'engineer': 1572 occurrences
- 'sales': 1258 occurrences
- 'customer': 964 occurrences
- 'senior': 963 occurrences
- 'service': 881 occurrences
- 'teacher': 774 occurrences
- 'english': 772 occurrences

Feature: description

Character Length Statistics:

- mean: 1218.0
- median: 1017.0
- std: 894.83
- min: 0
- max: 14907

Word Count Statistics:

- mean: 170.45
- median: 146.0
- std: 123.3
- min: 0
- max: 2115

Average Length - Fraudulent: 1154.83, Real: 1221.22

Top 10 Most Common Words:

- 'and': 175001 occurrences
- 'the': 98093 occurrences
- 'to': 96635 occurrences
- 'of': 67519 occurrences
- 'a': 62867 occurrences
- 'in': 53351 occurrences
- 'for': 46389 occurrences
- 'with': 41102 occurrences
- 'our': 30778 occurrences

- 'is': 28802 occurrences

Feature: requirements

Character Length Statistics:

- mean: 590.13
- median: 467.0
- std: 613.19
- min: 0
- max: 10864

Word Count Statistics:

- mean: 78.03
- median: 62.0
- std: 81.65
- min: 0
- max: 1645

Average Length - Fraudulent: 446.05, Real: 597.47

Top 10 Most Common Words:

- 'and': 82263 occurrences
- 'to': 46662 occurrences
- 'of': 35255 occurrences
- 'in': 31771 occurrences
- 'a': 30865 occurrences
- 'the': 26779 occurrences
- 'with': 25466 occurrences
- 'experience': 18204 occurrences
- 'or': 16450 occurrences
- 'for': 10701 occurrences

Feature: benefits

Character Length Statistics:

- mean: 208.9
- median: 45.0
- std: 337.08
- min: 0
- max: 4429

Word Count Statistics:

- mean: 29.99
- median: 6.0
- std: 49.76
- min: 0
- max: 694

Average Length - Fraudulent: 212.2, Real: 208.73

Top 10 Most Common Words:

- 'and': 22604 occurrences
- 'to': 13438 occurrences
- 'the': 13322 occurrences
- 'a': 13069 occurrences
- 'of': 11571 occurrences
- 'in': 7897 occurrences
- 'we': 7589 occurrences
- 'with': 6692 occurrences
- 'you': 5908 occurrences

- 'for': 5784 occurrences

Feature: company_profile

Character Length Statistics:

- mean: 620.9
- median: 570.0
- std: 567.45
- min: 0
- max: 6178

Word Count Statistics:

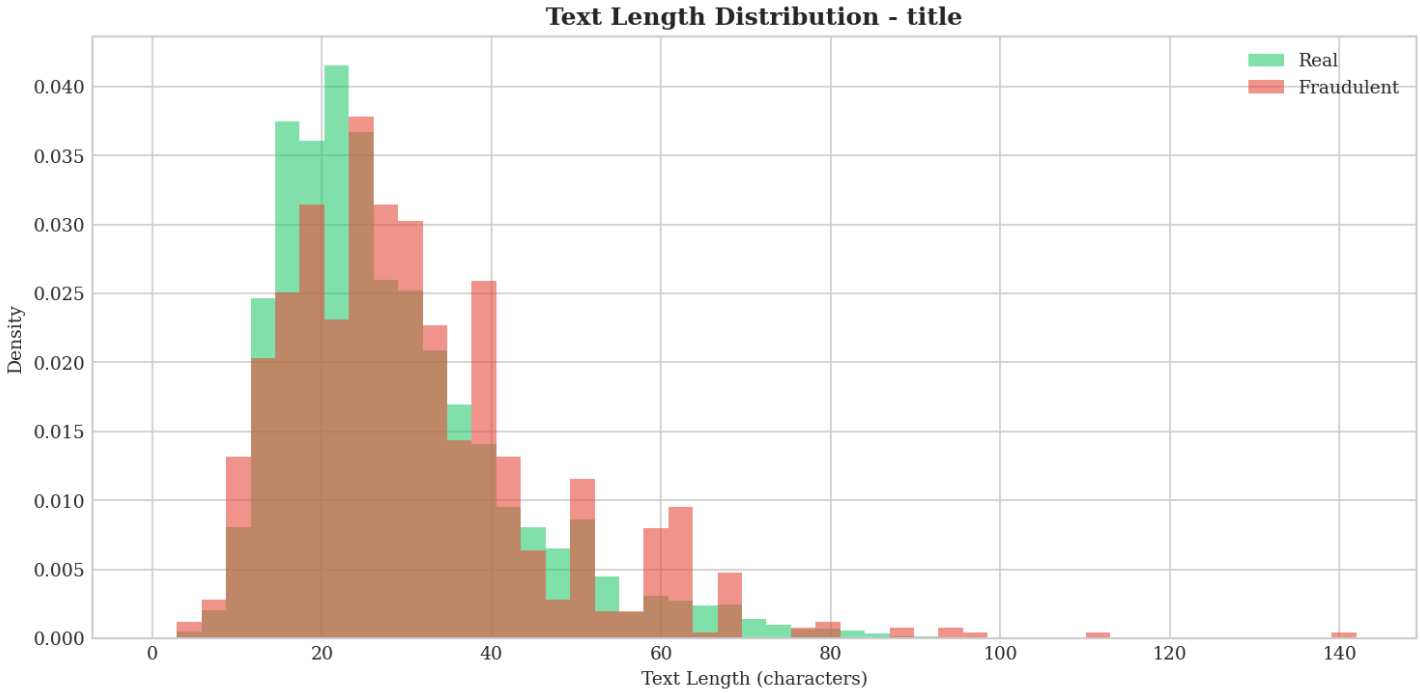
- mean: 92.55
- median: 83.0
- std: 85.49
- min: 0
- max: 885

Average Length - Fraudulent: 230.89, Real: 640.75

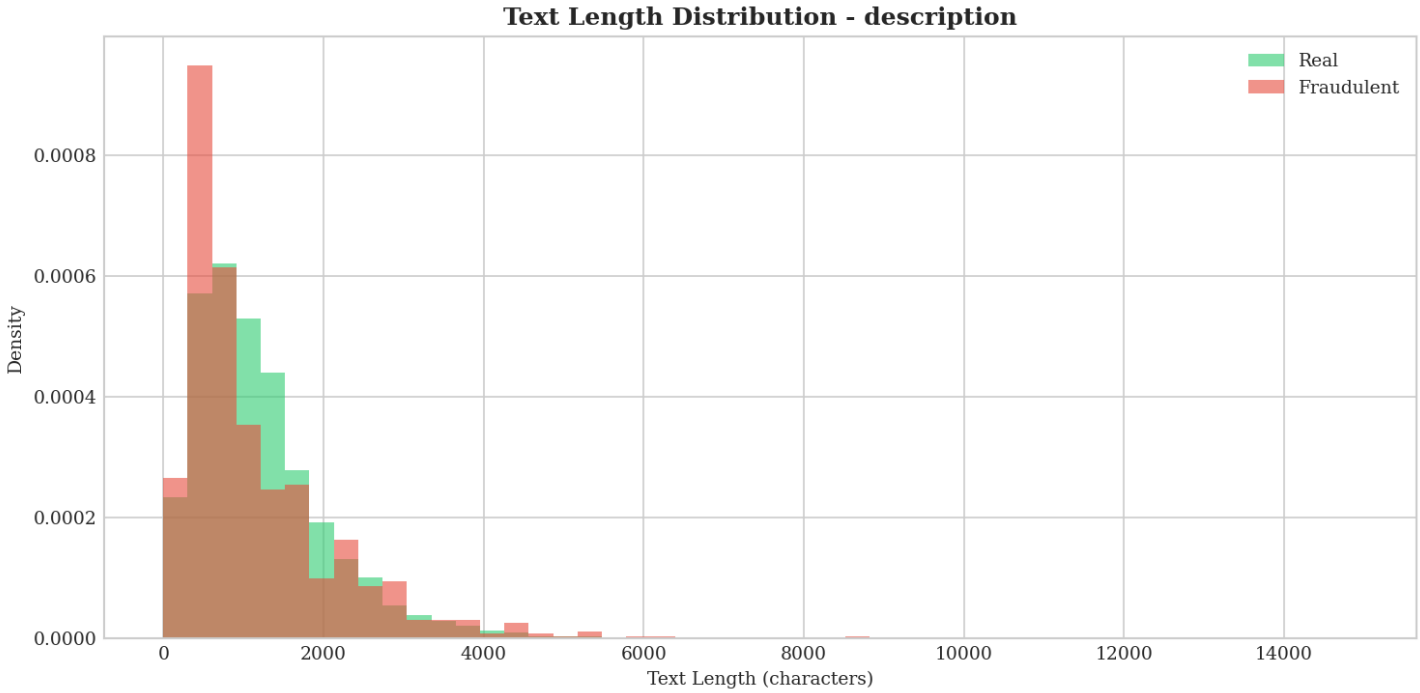
Top 10 Most Common Words:

- 'and': 79474 occurrences
- 'the': 61352 occurrences
- 'to': 48020 occurrences
- 'of': 38332 occurrences
- 'a': 35952 occurrences
- 'in': 32263 occurrences
- 'we': 27652 occurrences
- 'our': 25441 occurrences
- 'is': 22780 occurrences
- 'for': 20163 occurrences

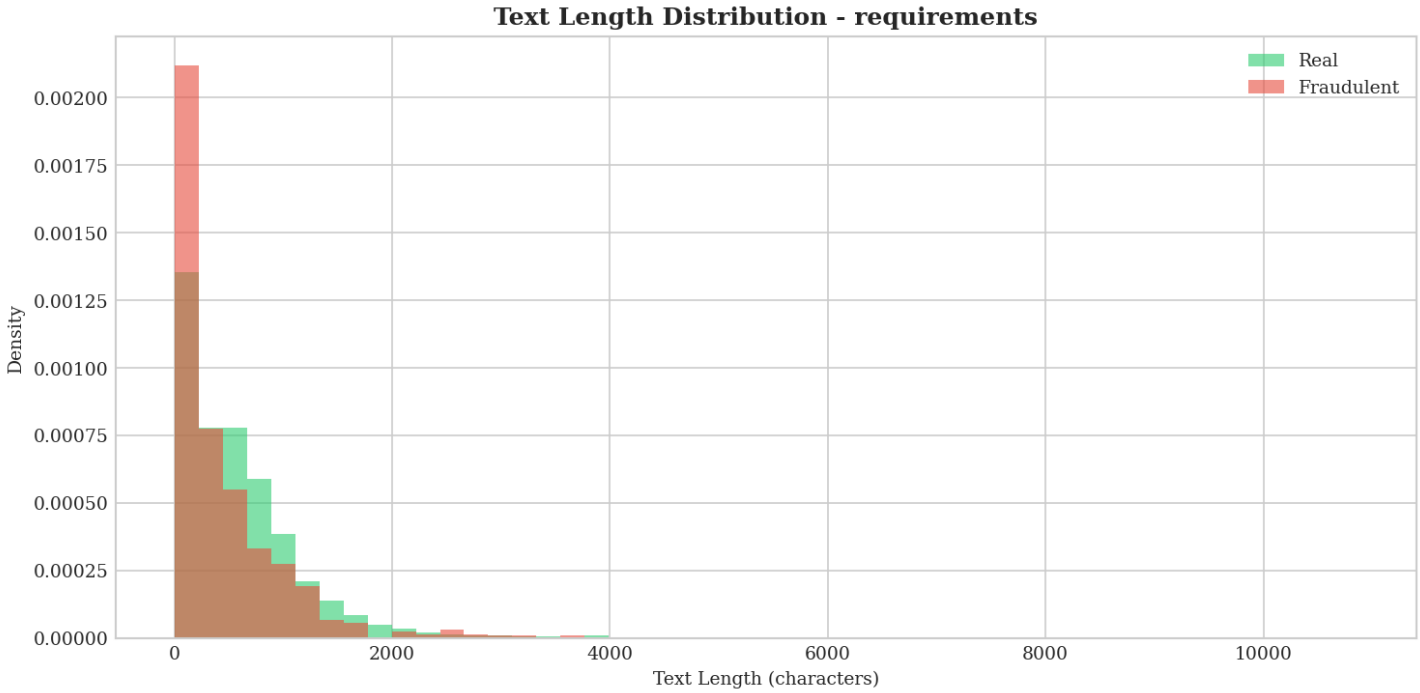
Text Length Distribution: title



Text Length Distribution: description



Text Length Distribution: requirements

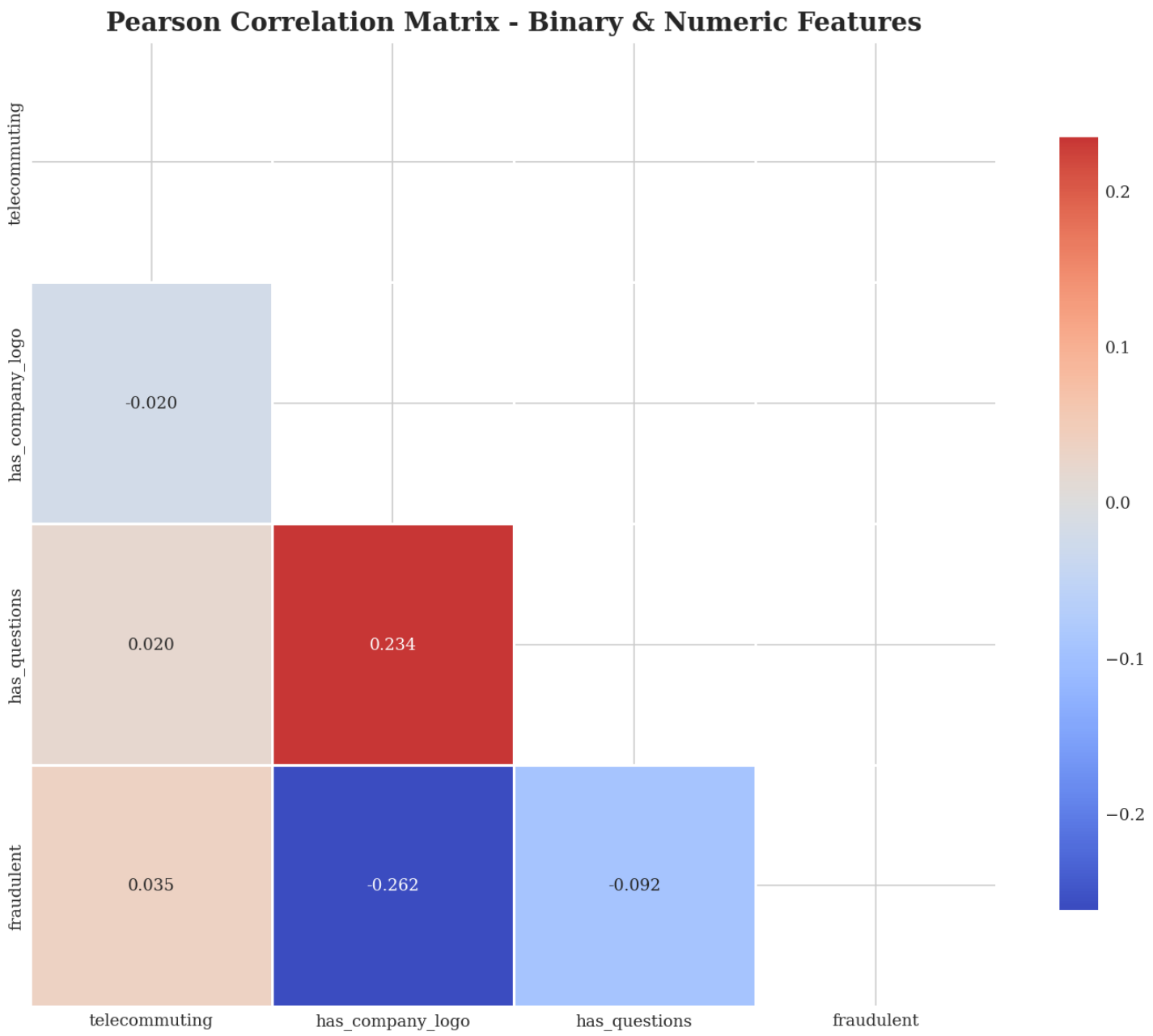


10. Comprehensive Correlation Analysis

All features ranked by correlation strength with fraudulent class:

Feature	Correlation	Method	P-Value	Significant
location	+0.68530	Cramér's V	N/A	*
department	+0.47731	Cramér's V	N/A	*
salary_range	+0.37223	Cramér's V	N/A	*
industry	+0.30131	Cramér's V	N/A	*
has_company_logo	-0.26197	Point-Biserial	2.21e-278	*
function	+0.17487	Cramér's V	N/A	*
required_education	+0.16106	Cramér's V	N/A	*
company_profile_length	-0.15506	Point-Biserial	1.25e-96	*
has_questions	-0.09163	Point-Biserial	1.20e-34	*
required_experience	+0.08417	Cramér's V	N/A	
employment_type	+0.07472	Cramér's V	N/A	
requirements_length	-0.05301	Point-Biserial	1.31e-12	*
title_length	+0.03474	Point-Biserial	3.38e-06	*
telecommuting	+0.03452	Point-Biserial	3.89e-06	*
description_length	-0.01593	Point-Biserial	3.32e-02	*
benefits_length	+0.00221	Point-Biserial	7.68e-01	

Correlation Matrix Heatmap

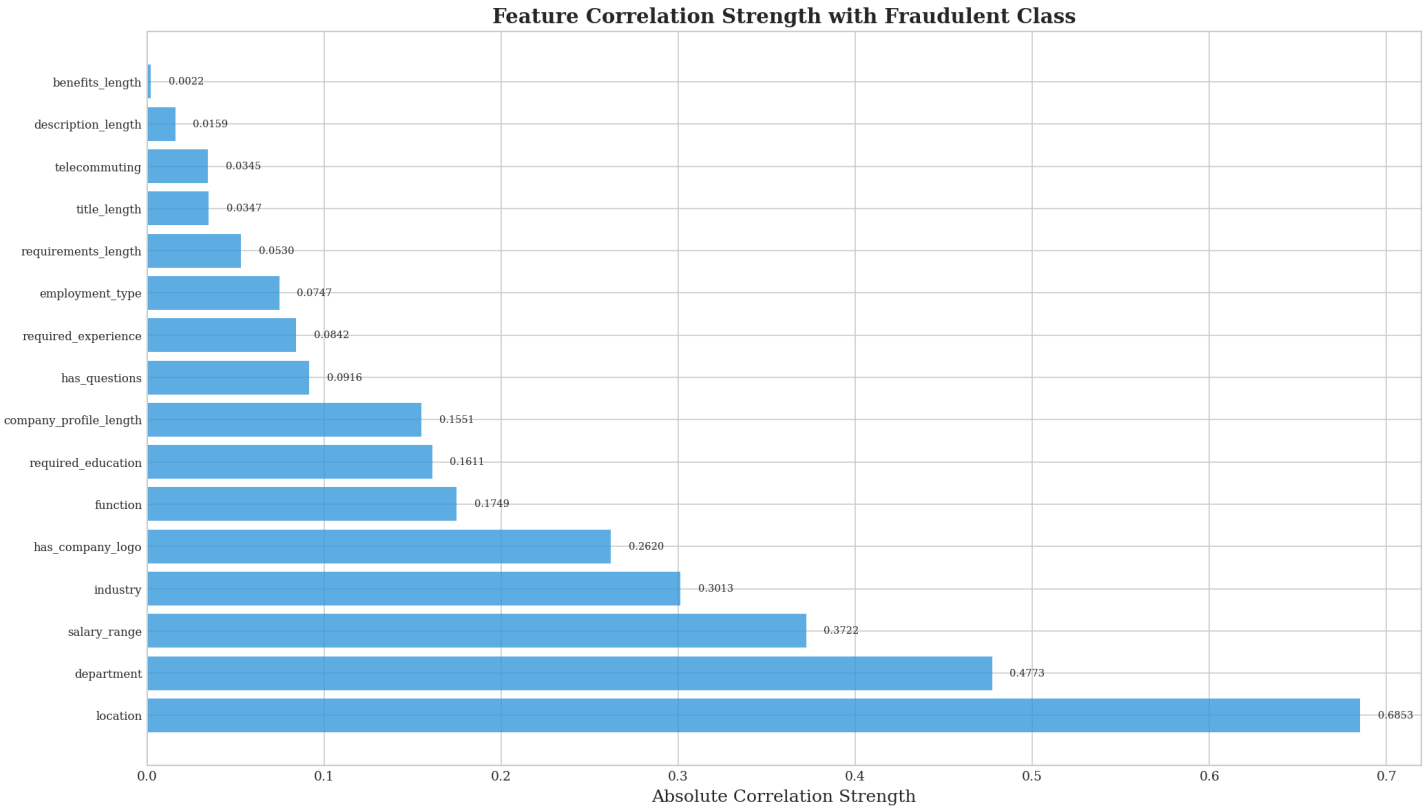


11. Feature Importance Ranking

Top 20 most predictive features for detecting fraudulent job postings:

Rank	Feature	Correlation	Method	Significance
1	location	0.68530	Cramér's V	Yes *
2	department	0.47731	Cramér's V	Yes *
3	salary_range	0.37223	Cramér's V	Yes *
4	industry	0.30131	Cramér's V	Yes *
5	has_company_logo	0.26197	Point-Biserial	Yes *
6	function	0.17487	Cramér's V	Yes *
7	required_education	0.16106	Cramér's V	Yes *
8	company_profile_length	0.15506	Point-Biserial	Yes *
9	has_questions	0.09163	Point-Biserial	Yes *
10	required_experience	0.08417	Cramér's V	No
11	employment_type	0.07472	Cramér's V	No
12	requirements_length	0.05301	Point-Biserial	Yes *
13	title_length	0.03474	Point-Biserial	Yes *
14	telecommuting	0.03452	Point-Biserial	Yes *
15	description_length	0.01593	Point-Biserial	Yes *
16	benefits_length	0.00221	Point-Biserial	No

Feature Importance Bar Chart



12. Statistical Summary

Summary of Statistical Tests:

Metric	Value
Total Features Tested	**16**
Statistically Significant Features	**13 (81.2%)**
Non-Significant Features	**3 (18.8%)**
Strongest Correlation	**location: +0.68530**
Weakest Correlation	**benefits_length: +0.00221**
Average Absolute Correlation	**0.18625**
Median Absolute Correlation	**0.12335**

13. Key Findings and Recommendations

KEY FINDINGS:

1. Dataset Composition:
   - The dataset contains 17,880 job postings with 4.84% fraudulent cases
   - Class imbalance exists which may require special handling in modeling

2. Most Predictive Features:

1. lo

   2. department (Correlation: +0.47731, Method: Cramér's V)
   3. salary_range (Correlation: +0.37223, Method: Cramér's V)
   4. industry (Correlation: +0.30131, Method: Cramér's V)
   5. has_company_logo (Correlation: -0.26197, Method: Point-Biserial)

3. Missing Data:
   - Several features have significant missing values that need imputation strategies
   - Missing data patterns may themselves be predictive of fraud

4. Text Features:
   - Text length and content analysis can provide valuable fraud detection signals
   - Fraudulent postings may have different linguistic patterns

5. Numeric Features:
   - Several numeric features show non-normal distributions
   - Outliers are present in some features and may need special handling
   - Statistical tests show significant differences between fraudulent and real postings

6. Recommendations:
   - Use all available features for model training, not just binary features
   - Apply appropriate encoding for categorical features
   - Consider text features using NLP techniques (TF-IDF, embeddings)
   - Address class imbalance using techniques like SMOTE or class weights
   - Validate model performance using appropriate metrics (F1, AUC-ROC)
   - Consider feature engineering based on the statistical insights